



COS1512 2019 October Exam

Introduction to programming II (University of South Africa)



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COS1512

October/November 2019

Introduction to Programming II

Duration 2 Hours

75 Marks

EXAMINERS

FIRST

SECOND

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Closed book examination

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This paper consists of 7 pages and 6 questions
Please make sure that you have all 7 pages with the 6 questions

Instructions:

- Answer all the questions
- Do all rough work in the answer book
- The mark for each question is given in brackets next to the question
- Please answer the questions in the correct order. If you want to do a question later, leave enough space
- Number your answers and label your rough work clearly
- Marks are awarded for part of an answer, so do whatever you are able to in each question

GOOD LUCK!

QUESTION 1**[7]**

1 1 Given the following C string variable declarations

```
char name[21], yourName[21],
```

Explain why the following statement is problematic and give the correct code to achieve the desired result. Note that there may be more than one aspect of the statement that is problematic.

```
if (yourName == name)
    studentName = name, (4)
```

1 2 Write statements to do the following

1 2 1 Declare a vector `v` with a base type of `int` and ten elements initialized to 0 (1)1 2 2 Add an element with the value 21 to the vector `v` (1)1 2 3 Change the size of vector `v` to 300 elements (1)**QUESTION 2****[5]**

Consider the following recursive function

```
string b (int n)
{
    string s;
    if (n%2 == 0)
        s = "0";
    else s = "1";
    if (n < 2)
        return s;
    return b(n/2) + s;
}
```

2 1 Identify the base case (1)

2 2 Identify the general case (1)

2 3 Function `b()` is called in the following program fragment

```
string result = b(8);
cout << "8 converted by function b() is " << result;
```

2 3 1 What is the output produced by this program fragment? (1)

2 3 2 What does function `b()` calculate? (2)

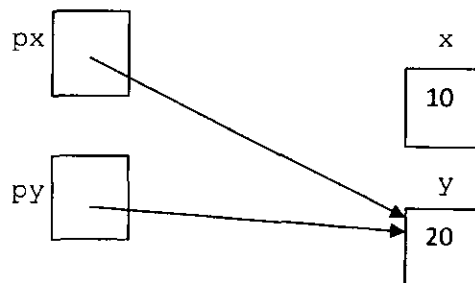
QUESTION 3**[7]**

3 1 Consider the following declarations and answer the questions that follow

```
int *px;
int *py;
int x = 10;
int y = 20;
```

3 1 1 Provide C++ statements so that pointer variables `px` and `py` point to variables `x` and `y` respectively (2)

3 1 2 Write a C++ code segment to recreate the pointer structure given below by using the declarations given above (1)



3 1 3 What will happen when the statement

```
delete px,
```

is executed?

(1)

3 2 Write C++ statement(s) to create a dynamic array `dArray` that can hold `arraySize` integers (3)

QUESTION 4**[33]**

Note: Read through the whole of Question 4 below before you attempt to answer the questions that follow. Also, provide **only** the member functions requested and required by the question.

The Exams department keeps a file with a list of all the modules in which exams are written. For each module, the module code and module name are stored in the file `Exams.dat`.

Once the exam timetable has been determined, the file `Exams.dat` needs to be updated with the number of students who will write the exam as well as the date on which the exam will be written. However, every exam period, the same exams are written. The Exams department, therefore, wants to keep the file `Exams.dat` intact. They require a program that will copy the contents of the file `Exams.dat` to another file called `ExamsDetail.dat` in which the number of students who will write the exam as well as the date on which the exam will be written, have been added to the module code and module name for each exam. This will create a record for each exam in file `ExamsDetail.dat`.

TURN OVER

Complete the program below, using separate compilation, to assist the Exams Department

The program uses a class `Exam`. This class has four member variables

- `mCode`, a string for the module code
- `mName`, a string for the module name
- `nrStudents`, an integer value to indicate the number of students who will write the exam
- `examDate`, a string for the date on which the exam will be written

In addition, the class `Exam` has the following member functions

- A default constructor that initialises all the strings to empty strings, and the integer member variable to 0
- An overloaded constructor to initialise member variables to specified values
- A destructor
- An accessor `getCode()`
- A mutator `setDate()`
- An overloaded stream extraction operator `>>` implemented as a `friend` function, to read an `Exam` object from a file
- An overloaded stream insertion operator `<<` implemented as a `friend` function, to write an `Exam` object to a file

The application program reads the module code and module name for each exam from the file `Exams.dat`, obtains the number of students who will write the exam and the date on which the exam will be written, from the user, and then writes that information to the file `ExamsDetail.dat`.

An example of the input (in bold) for the first module in file `Exams.dat` is shown below

```
Please enter the number of students for the module COS1234 Programming: 583
Please enter the exam date for COS1234 Programming: 12122019
```

- 4.1 Create the header file `Exam.h` for the class specification (9)
- 4.2 Create the implementation file `Exam.cpp` for the class implementation including all the `friend` functions (10)
- 4.3 Complete the application program (`main()`) below to create the file `ExamsDetail.dat`. Write down only the number and the missing statement(s). Be sure to use appropriate member functions in the required statements (9)

```
#include <iostream>
_____ 1 _____ //1.Include files needed
using namespace std;

int main()
{
    _____ 2 _____ //2.Declare input file, open file Exams.dat
    // and check that file exists (2 marks)

    _____ 3 _____ //3.Declare output file, open file
    // ExamsDetail.dat and check that file exists (2 marks)
```

TURN OVER

```

string code, name, date,
int number;

while ((_____4_____)//4.Read the module code and module
// name from file Exams.dat
{
    cout << "Please enter the number of students for the module "
        << code << " " << name << ". ",
    cin >> number;
    cout << "Please enter the exam date for " <<code << " "
        << name << ". "; //
    cin >> date,
        _____5_____/5 Instantiate an object anExam using the
//overloaded constructor to initialise the object with the values
//obtained
        _____6_____/6.Output object anExam to file
//ExamsDetail.dat
}

_____7_____/7.Close files

return 0,
}

```

- 4.4 Complete the code fragment below to update the exam date for a module. The internal filename `inputfile` has been associated with the input file, and has been opened successfully. This file is used as input file in the `while` loop. The updated exam object as well as unaltered exam objects are copied to an output file with an internal filename `outputfile`. This file has also been opened successfully. (5)

```

string newdate;
cout << "For which module do you need to update the exam details?"
    << endl;
cin >> code;
cout << "Enter new exam date: " ;
cin >> newdate;

_____1_____/1.Instantiate an object aExam for an exam using
//the default constructor
while ((_____2_____)//2.Read an object aExam from the
// input file
{
    if (_____3_____)//3.Compare variable code with the
//member variable mCode in object aExam
    {
        _____4_____/4.Set member variable examDate in
//object aExam to the new date for the exam
    }
    _____5_____/5.Output object aExam to the output file
}

```

TURN OVER

QUESTION 5**[14]**

Consider the class specification (interface) for the class `Circle` below

```
const double PI = 3.1415927,

class Circle
{
    public:
        Circle (string name, double rad),
        double area () const;
        string getShapeName() const;
    private:
        double radius,
        string shapeName;
};
```

- 5.1 Define a new class `Cylinder` that inherits the functionality of the class `Circle`. The class `Cylinder` has an additional member variable, `height`, and an additional member function, `volume()` to calculate the volume of the cylinder. The class `Cylinder` should override function `area()` to calculate the area of the cylinder.

Provide only the interface of class `Cylinder` in a header file. The header file should contain compiler directives to prevent multiple definitions. Assume that the interface of class `Circle` is contained in an interface file called `Circle.h`. (7)

- 5.2 Implement the overloaded constructor for the class `Cylinder` by invoking the base class constructor. (3)

- 5.3 Consider the following implementation to calculate the volume of a cylinder.

```
double Cylinder volume() const
{
    return PI * radius * radius * height;
}
```

Compiling this implementation produces an error message. Explain why and provide two ways in which this problem can be resolved. (3)

- 5.4 The area of the base of the cylinder is the same as that of the class `Circle` from which the class `Cylinder` is derived. Write a statement to display the area of the base of a `Cylinder` object `myCylinder` by calling the `area()` member function in class `Circle`. (1)

QUESTION 6**[9]**

A list is defined as a set of values of the same type. The interface for a class `ListType` that uses an array to represent and process a list of integer values appears below.

TURN OVER

```

class ListType
{
public
    ListType (),                      //constructor
    bool isEmpty() const,
    bool isFull() const,
    int search(int searchItem) const, // returns the position of
                                      // searchItem if it is
                                      // found in the list, else
                                      // -1 is returned

    void insert(int newElement),
    void destroyList();
private.
    int list[100]; //array to hold the list elements
    int length     //number of elements in list

};

```

- 6 1 Write the interface for a generic class `ListType` that can store a list of values of any type, e.g. a list of chars or objects of a user defined class. Provide only the interface (4)
- 6 2 Implement the `insert()` member function of the template class `ListType`. (4)
- 6 3 Provide a declaration for a `ListType` of chars (1)